

$$\mathcal{A}_{\varepsilon,\ell,p}^L(\mathbf{r}) = \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\varepsilon,\ell,p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z}$$